

The original terminal arithmetic response Separate denominator rates and complex current coherence

Split-Zero research programme

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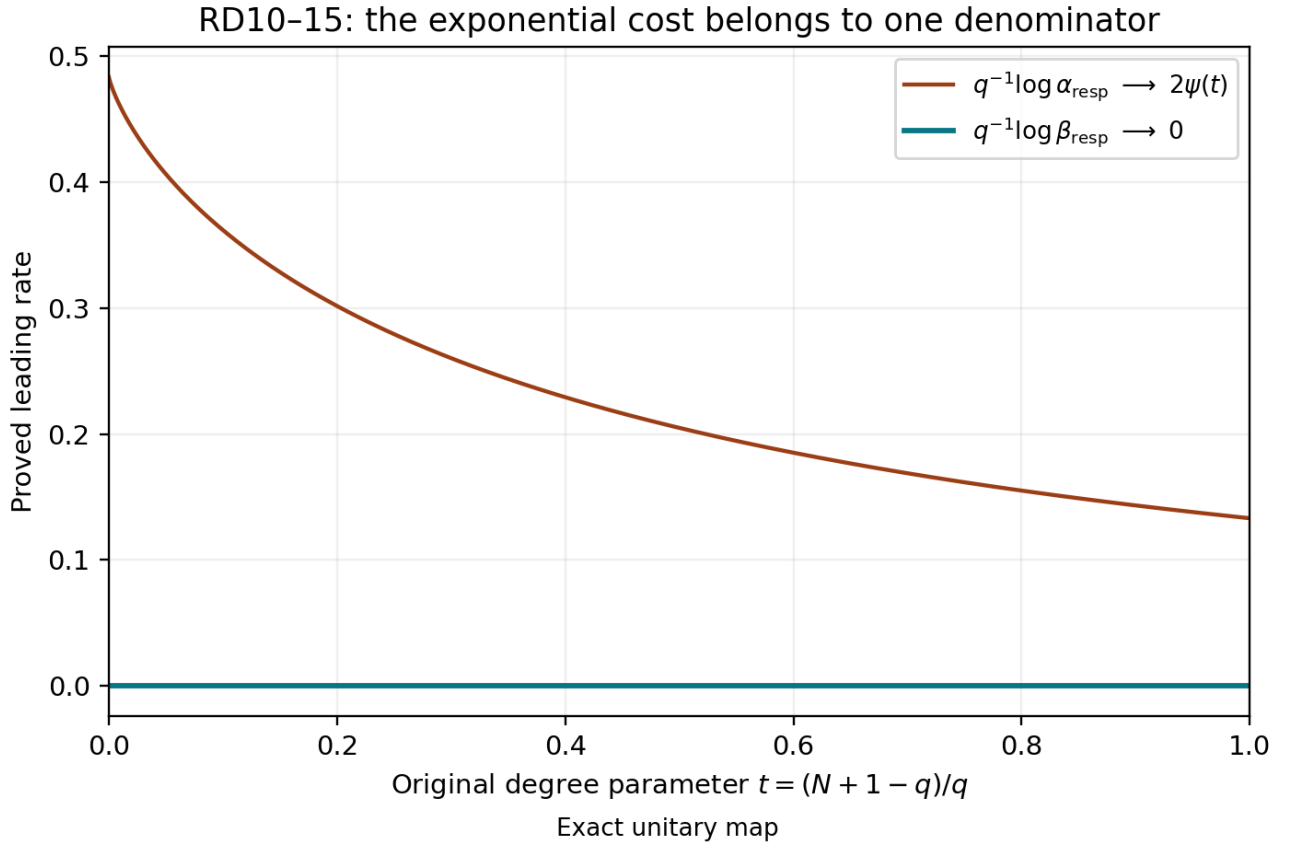
The two boundary denominators of the original arithmetic eigenclass have different growth rates. The first carries the complete exponential amplification; the second has logarithmic size at most $O(k + \log q)$. The actual observation responses retain all their complex phases and complete kernel components.

On the established conductor period domain, the observed terminal class lies exponentially close to a precisely defined complex hyperplane. This controls the complex amplitudes in the current's positive and negative eigenspaces, improving the preceding comparison of their squared magnitudes. The diagonal trace correction to its current is absolutely exponentially small. The remaining sign belongs to the retained imaginary cross-coordinate and is not assigned by these bounds.

All original sources, cutoffs, maps and metrics are retained. The preceding complete proof collection is included unchanged in the accompanying recoverable source bank, followed by these two complete new proofs and their reproducible auxiliary checks.

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$$\begin{aligned} c_g &= \langle g, \hat{y} \rangle \\ c_h &= \langle h, \hat{y} \rangle \end{aligned}$$



$$\begin{aligned} z_+ &= iA_*c_g + B_*c_h \\ z_- &= -iB_*c_g + A_*c_h \end{aligned}$$

$$A_*z_+ - B_*z_- = ic_g, \quad |c_g| \leq \eta_N$$

$$|z_+ - z_-|^2 \leq (\sqrt{2}\eta_N + \delta_N^{\text{tr}})^2$$

$$\text{TR18: coherence deficit} \leq \exp[-2q\psi(t) + O(k \log q)]$$

Figure 1: Upper: the exact leading profiles in RD14, sampled from the retained elliptic formulas; these curves are limiting rates, not finite native-period data. The errors remain those stated in RD14-15. Lower: the exact complex-coordinate map TR14-18 in the original metric. A_*, B_* are positive and have squared sum one. No phase angle is assigned at zero amplitude; the squared difference is valid there as well. The complete proof retains every zero eigendirection. Human source for the elliptic formulas: B. C. Carlson, DLMF 19.5 and 19.8; the programme-specific receiving maps and estimates are proved in full below.

The two original response denominators have different evaluated growth rates

Result SZ-20260922-027, root derivation, 22 September 2026. This is a continuation of the sealed 026 edition, which remains unchanged. All norms and adjoints below use the original arithmetic quotient metric. The finite argument determines the two denominator amplification rates separately and bounds the actual response ratios. It retains their complex phases and does not assign the residual marked-current sign.

The two denominators are inner products of the fixed terminal eigenclass with the last two source-polynomial classes. Each amplification compares the product of their squared norms with their squared overlap. A large amplification means a small overlap relative to those norms. Earlier work evaluated the product of the two amplifications; it did not identify which overlap was small. The eigenclass equation now proves that the first carries the exponential cost. This matters because the same denominators occur in the two complex response ratios used to calculate the observed, hidden and mixed arithmetic currents.

1. Original objects and exact phase dictionary

Retain the stipulated simple quartet, original admitted period and arithmetic unit, source measure, all invariant columns, and every source-minimum fibre. Write

$$k \equiv 1 \pmod{4}, \quad k \geq 17, \quad q = (k+1)^2, \quad q_0 = (k-7)^2, \quad q-1 \leq N \leq 2q,$$

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E = \mathbb{C}[y]/Q_k, \quad S = k/2 + iy,$$

$$M[p] = [yp], \quad \mathcal{A} = kI/2 + iM, \quad \omega = k\gamma - ik\delta, \quad d = k\delta > 0. \quad (\text{RD1})$$

The full metric G_N , onto map $\Lambda : E \rightarrow B$, kernel $K = \ker \Lambda$, and original minimum section are

$$Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_N = G_N^{-1} \Lambda^* Q_{B,N}, \quad P = L_N \Lambda, \quad Q = I - P. \quad (\text{RD2})$$

Thus P, Q are the actual complementary G_N -orthogonal projections. The notation Q for the projection is distinct from the polynomial Q_k .

Let p_n^y be the original real monic orthogonal polynomials, with full squared source norms ω_n , and put

$$b_n^y = [p_n^y], \quad E_N = \|b_N^y\|_{G_N}^2, \quad F_N = \|b_{N+1}^y\|_{G_N}^2,$$

$$e = b_N^y / \sqrt{E_N}, \quad f = b_{N+1}^y / \sqrt{F_N}, \quad \epsilon = \sqrt{E_N F_N} / \omega_N. \quad (\text{RD3})$$

These unit vectors only express the already defined NCE20 ratios; no source measure or metric is changed. The original full action has the exact decomposition

$$M = C + \epsilon f e^{\dagger G_N}, \quad C^{\dagger G_N} = C, \quad \langle e, f \rangle_{G_N} = 0, \quad \|C\|_{G_N} \leq c_N := 2(N+1)\sqrt{u_k/\ell_k}. \quad (\text{RD4})$$

This is OCP5–7 and RC22–25 of the complete 026 proofs. Its coefficient proof is worth retaining: multiplication by real y , compressed to the complete attained source section, is selfadjoint; its only omitted polynomial is the leading coefficient times p_{N+1}^y . The leading coefficient is $(b_N^y)^* G_N / \omega_N$. The two outgoing classes have opposite parities in the even source and even relation ideal. This proves RD4, including its rank-one orientation and orthogonality. The bound for C is the full Gamma multiplication bound transported by the two original source inequalities through degree $2q+2$.

Let $x = v_+$ be the original terminal cardinal, so $Mx = \omega x$. Put

$$A = \langle e, x \rangle_{G_N}, \quad B_f = \langle f, x \rangle_{G_N}, \quad E_x = \|x\|_{G_N}^2, \quad D_N = |\omega| + c_N. \quad (\text{RD5})$$

The subscript on B_f distinguishes this scalar from the observation space. Inner products are conjugate linear in their first argument.

The physical monic classes are exactly $b_n^S = i^n b_n^y$, under the pullback $S = k/2 + iy$. Therefore the original NCE1 Gram entries obey

$$F_{13}^S = (-i)^N \sqrt{E_N} A, \quad F_{23}^S = (-i)^{N+1} \sqrt{F_N} B_f. \quad (\text{RD6})$$

This is the complete diagonal unit-phase change between the two boundary rows. It gives the actual current coefficient and the original NCE20 amplifications:

$$\mathfrak{c} = \frac{\overline{F_{13}^S} F_{23}^S}{\omega_N E_x} = -i\epsilon \frac{\overline{A} B_f}{E_x}, \quad \alpha_{\text{resp}} = \frac{E_x}{|A|^2}, \quad \beta_{\text{resp}} = \frac{E_x}{|B_f|^2}. \quad (\text{RD7})$$

Equations RD6–7 prove the typed correspondence to the earlier three-column calculation, including its imaginary phase. That calculation is retained at NCE1–21, especially NCE9 and NCE20–21.

2. Separate finite bounds for both denominators

The eigenclass equation and RD4 give

$$\epsilon f A = (\omega I - C)x, \quad \epsilon |A| \leq D_N \sqrt{E_x}. \quad (\text{RD8})$$

Taking the inner product with x in RD4, the C -term is real. Since $\Im \omega = -d$,

$$\epsilon \Im(\overline{B_f} A) = -d E_x, \quad \Re \mathfrak{c} = d, \quad \epsilon |A B_f| \geq d E_x. \quad (\text{RD9})$$

In particular both denominators are nonzero for every original cutoff, independently of a nonvanishing period minor. Since $|A|, |B_f| \leq \sqrt{E_x}$, RD8–9 prove the complete finite bounds

$$\boxed{\max\{1, \epsilon^2/D_N^2\} \leq \alpha_{\text{resp}} \leq \epsilon^2/d^2, \quad 1 \leq \beta_{\text{resp}} \leq D_N^2/d^2.} \quad (\text{RD10})$$

For the last inequality, use RD8 in RD9 to obtain $|B_f| \geq d\sqrt{E_x}/D_N$. For the upper bound on α_{resp} , use $|B_f| \leq \sqrt{E_x}$ in RD9. No estimate of a determinant has been substituted for an individual column. The exact product remains

$$\alpha_{\text{resp}} \beta_{\text{resp}} = \epsilon^2/|\mathfrak{c}|^2, \quad (\text{RD11})$$

as in NCE21, but RD10 now evaluates which factor carries the exponential scale.

Retain the literal source-comparison constants through degree $2q + 2$. FW7 and RC22 give $\log(u_k/\ell_k) = O_h(k + \log q)$. The full monic-profile estimate and RC25 give

$$\log \epsilon = q\psi(t_N) + O_{h,\varpi}(k \log q), \quad t_N = (N + 1 - q)/q. \quad (\text{RD12})$$

Because $d = k\delta$, $|\omega| = k\sqrt{\delta^2 + \gamma^2}$, and $N \leq 2q$,

$$0 \leq \log(D_N/d) = O_h(k + \log q). \quad (\text{RD13})$$

Indeed $u_k/\ell_k \geq 1$, and each summand of D_N/d is bounded by an original fixed constant times $q\sqrt{u_k/\ell_k}/k$. Its logarithm has the claimed bound. Combining RD10–13 proves

$$\boxed{\log \alpha_{\text{resp}} = 2q\psi(t_N) + O_{h,\varpi}(k \log q), \quad 0 \leq \log \beta_{\text{resp}} = O_h(k + \log q).} \quad (\text{RD14})$$

The conclusion is uniform on the complete original cutoff window. It is an evaluation of the separate leading rates; it does not declare β_{resp} constant or determine its smaller terms.

The four original cutoffs have $t_N = 0, 1/q, 1, 1 + 1/q$ and signs $+, +, -, -$. The retained profile satisfies $\psi(t) - \psi(0) = O(t|\log t|)$ at zero and has a bounded derivative near one. Thus RD14 gives

$$\boxed{\mathcal{R} \log \alpha_{\text{resp}} = 4[\psi(0) - \psi(1)]q + O_{h,\varpi}(k \log q) = 2C_\partial q + O_{h,\varpi}(k \log q),}$$

$$\mathcal{R} \log \beta_{\text{resp}} = O_h(k + \log q). \quad (\text{RD15})$$

Here $C_\partial = 2[\psi(0) - \psi(1)]$ is the exact previously evaluated coefficient. The endpoint estimates follow directly from RC1–10: $t = z + O(z^2)$, while the only unbounded derivative at zero is the displayed $\log z$ term. The smooth elliptic parameter at $t = 1$ lies strictly below one. Hence no omitted endpoint is absorbed without a bound.

3. The actual responses also have finite bounds

Keep the actual kernel projector Q , and define its three original energies

$$\alpha_K = \|Qe\|^2, \quad \beta_K = \|Qf\|^2, \quad \theta_K = \|Qx\|^2/E_x. \quad (\text{RD16})$$

All lie in $[0, 1]$. These quantities do not replace any kernel columns. The unchanged physical response ratios are

$$U = \frac{\langle e, Qx \rangle}{A}, \quad V = \frac{\langle f, Qx \rangle}{B_f}. \quad (\text{RD17})$$

The same unit phase in numerator and denominator cancels by RD6, proving that RD17 is exactly NCE14. Orthogonality of Q and Cauchy–Schwarz give

$$|\langle e, Qx \rangle|^2 \leq \alpha_K \theta_K E_x, \quad |\langle f, Qx \rangle|^2 \leq \beta_K \theta_K E_x.$$

Therefore RD10 gives

$$\boxed{|U| \leq \frac{\epsilon \sqrt{\alpha_K \theta_K}}{d}, \quad |V| \leq \frac{D_N \sqrt{\beta_K \theta_K}}{d}}. \quad (\text{RD18})$$

Every original cross term remains in these ratios. The sharper RC25 bound $\epsilon \sqrt{\alpha_K} \leq \exp[O_{h,\varpi}(k \log q)]$ and RD13 now imply

$$\log(1 + |U|) = O_{h,\varpi}(k \log q), \quad \log(1 + |V|) = O_h(k + \log q). \quad (\text{RD19})$$

Consequently each of the exact NCE15 products

$$P_K = \overline{U}V, \quad P_B = (1 - \overline{U})(1 - V), \quad P_\times = \overline{U} + V - 2\overline{U}V \quad (\text{RD20})$$

has $\log(1 + |P_X|) = O_{h,\varpi}(k \log q)$. This follows by the triangle inequality and RD19, without removing any of the three terms. Their exact currents remain

$$\Phi_X/(2E_x) = \Re(cP_X), \quad P_K + P_B + P_\times = 1. \quad (\text{RD21})$$

The full source sign $\Re c = d > 0$ is retained. The complex factors in RD20 still determine the separate signs. In particular, RD19 is not an arithmetic-scale error estimate: a subexponential factor can still amplify an error of polynomial size. The next signed calculation must retain the actual products, rather than discarding them because their exponential rate is zero.

4. Receivers and provenance

RD10 and RD14 strengthen the formerly product-only statement NCE21. RD18–21 carry those individual bounds into the same original EC37–41/NCE14–17 responses. They hold for the stipulated source eigenclass without the additional large-period corner guard used for RC40–44. Any later division by the observed norm retains that guard or another proved norm estimate.

The complete preceding programme proofs are NCE1–21 at the pinned link above; OCP1–14 and RC22–27 in the sealed 026 edition; and the full source inequality FW7. The new argument RD8–21 is given in full here. Human inputs to the retained source and monic estimates are R. A. Askey and R. Roy, DLMF 5.7.6; T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, DLMF 18.22.8 and 18.23.7; and B. C. Carlson for the elliptic series. Their original equation sources and actual prior reading records remain in the preceding source bank; they are not claimed as authors of the programme-specific denominator estimates.

Complex phase coherence of the original terminal current

22 September 2026. This is a continuation of RC35-44 and CE1-29. The exact original terminal class has exponentially balanced current-sign weights. Here the eigenclass equation controls their complex amplitudes, with a doubled exponential rate for the phase-coherence deficit. All estimates use the original observation projection; the argument does not optimize over substitute projections.

1. Original object and proved domain

Keep $k \equiv 1 \pmod{4}$, $k \geq 17$, $q = (k+1)^2$, $q_0 = (k-7)^2$, the original simple quartet $0 < \delta < 1/2, \gamma > 2$, admitted period and branch. In the physical coordinate $S = k/2 + iy$, the coefficient algebra, multiplication and terminal cardinal are

$$E = \mathbb{C}[y]/Q_k, \quad M[p] = [yp], \quad x = v_+(S) = \frac{\chi_k(S)}{(S - k\rho)\chi'_k(k\rho)}, \quad \rho = \frac{1}{2} + \delta + i\gamma, \quad Mx = \omega x, \quad \omega = k\gamma - ik\delta. \quad (\text{TR1})$$

For every original cutoff $q-1 \leq N \leq 2q$, retain the entire source metric G_N , original onto map Λ , full invariant kernel $K = \ker \Lambda$, and

$$Q_B = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L = G_N^{-1} \Lambda^* Q_B, \quad P = L\Lambda. \quad (\text{TR2})$$

The adjoint is in the stated metric, with inner products conjugate-linear in the first argument. CE2-3 prove that P is the G_N -orthogonal projection onto $K^\perp$; $\Lambda : (K^\perp, G_N) \rightarrow (B, Q_B)$ is an isometry with inverse L . Consequently every calculation below on $y = Px$ is a calculation of the original observed class Λx , with its actual minimum norm.

The terminal nonvanishing and quantitative norm comparison are used precisely on the period domain of RC37-42: $\gamma \geq 1000\delta$, the retained five-orbit requirement, and $|\varpi| \geq R_{\text{corner}}$, with the actual finite R_{corner} of RC37-38. Its proof uses the original coefficient $C_{Av_+} = a_{88}v_-$, not a selected replacement column. Write

$$\theta = \frac{\|Px\|_{G_N}^2}{\|x\|_{G_N}^2}, \quad \theta_K = 1 - \theta, \quad \theta \geq \mathfrak{l}_{k,N} > 0, \quad -\log \mathfrak{l}_{k,N} = O_{h,\varpi}(k \log(q+2)). \quad (\text{TR3})$$

The complete finite formula for $\mathfrak{l}_{k,N}$ and its conductor/reverse-lift proof are RC40-42; no other period domain is inferred here.

Retain the full even-source monic classes b_N, b_{N+1} and their norms E_N, F_N, ω_N . Put

$$e = b_N/\sqrt{E_N}, \quad f = b_{N+1}/\sqrt{F_N}, \quad \epsilon = \sqrt{E_N F_N}/\omega_N, \quad (\text{TR4})$$

$$M = C + \epsilon f e^\dagger, \quad C = C^\dagger, \quad e \perp f, \quad \|e\| = \|f\| = 1.$$

This is the exact original action CE5/OC5-7. The degree- $N+1$ class is represented by its complete degree- N minimum, so the cutoff has not changed. Set

$$c = 2(N+1)\sqrt{u_0/\ell_0}, \quad \|C\| \leq c, \quad D = |\omega| + c, \quad u = Pe, \quad v = Pf, \quad a = \|u\|^2, \quad d = \|v\|^2, \quad r = \langle u, v \rangle, \quad (\text{TR5})$$

$$C = \begin{pmatrix} a & r \\ \bar{r} & d \end{pmatrix}, \quad \mathfrak{d} = ad - |r|^2, \quad \alpha = \|(I - P)e\|^2 = 1 - a.$$

The notation here is exactly $u_0 = u_k, \ell_0 = \ell_k$, the original complete source constants of RC22/RD4 through degree $2q+2$; thus the actual source-comparison ratio is retained. On the eventual finite guards proved in the conductor calculation,

$$0 \prec \mathcal{C} \preceq I_2, \quad \mathfrak{d} \geq \Theta_N > 0, \quad |r|^2 \leq \alpha(1-d) \leq \alpha. \quad (\text{TR6})$$

For the last inequality, $e \perp f$ gives $r = -\langle (I - P)e, (I - P)f \rangle$, then apply Cauchy-Schwarz to these actual hidden vectors. Since the larger eigenvalue of $\mathcal{C}$ is at most one, its smaller eigenvalue is at least its determinant; consequently $a, d \geq \Theta_N$.

2. The terminal equation constrains a complex coordinate

Let $A = \langle e, x \rangle$, $B = \langle f, x \rangle$. From TR1 and TR4,

$$\epsilon A f = (\omega - C)x, \quad \epsilon |A| \leq D \|x\|. \quad (\text{TR7})$$

This is a vector equation before observation. After the complete original projection,

$$\langle e, Px \rangle = A - \langle (I - P)e, (I - P)x \rangle.$$

Hence

$$\frac{|\langle e, Px \rangle|}{\|Px\|} \leq \frac{D/\epsilon + \sqrt{\alpha\theta_K}}{\sqrt{\theta}}. \quad (\text{TR8})$$

The conductor lower bound TR3 is what makes the denominator quantitative. It is not omitted.

Define the exact orthonormal pair in the original observed space by

$$g = \frac{u}{\sqrt{a}}, \quad h = \frac{v - (r/a)u}{\sqrt{\delta/a}}. \quad (\text{TR9})$$

The subtraction is the actual projection of v on u . Direct multiplication with TR5 proves $\langle g, h \rangle = 0$ and both norms equal one. In particular no cross phase has been discarded. Let

$$\hat{y} = \frac{Px}{\|Px\|}, \quad c_g = \langle g, \hat{y} \rangle, \quad c_h = \langle h, \hat{y} \rangle, \quad y_0 = \hat{y} - c_g g - c_h h.$$

Then y_0 is orthogonal to both g, h , and

$$|c_g| \leq \eta_N := \min \left\{ 1, \frac{D/\epsilon + \sqrt{\alpha\theta_K}}{\sqrt{a\theta}} \right\}. \quad (\text{TR10})$$

Indeed $P^\dagger = P$ and $P\hat{y} = \hat{y}$, so $\langle g, \hat{y} \rangle = \langle e, Px \rangle / (\sqrt{a}\|Px\|)$; apply TR8. TR10 controls the entire complex coordinate, not only its imaginary part.

Using $\theta \geq \iota_{k,N}$, $a \geq \Theta_N$, and the completed source estimates RC25,

$$\begin{aligned} \log \epsilon &= q\psi(t_N) + O_{h,\varpi}(k \log q), & -\log \Theta_N &= O_{h,\varpi}(k \log q), \\ \alpha &\leq e^{-2q\psi(t_N) + O_{h,\varpi}(k \log q)}, & \log D &= O_h(k + \log q). \end{aligned}$$

$$t_N = (N + 1 - q)/q,$$

we obtain, uniformly on all original cutoffs and the stated period domain,

$$\eta_N \leq \exp[-q\psi(t_N) + O_{h,\varpi}(k \log q)]. \quad (\text{TR11})$$

The profile ψ is the original monic profile in RC8/OCP31, not the heat profile J ; their exact relation is RC7-10.

There is a geometric consequence in the original metric. If $\mathcal{H}_N = \{b \in K^\perp : \langle Pe, b \rangle = 0\}$, then

$$\text{dist}_{G_N}(\hat{y}, \mathcal{H}_N) = |c_g| \leq \eta_N. \quad (\text{TR12})$$

This is a complex hyperplane, with exactly defined orthogonal projection $I - gg^\dagger$ on $K^\perp$. It contains the original zero-current line generated by h and the entire zero eigenspace of the current. Equation TR12 does not assert that $\hat{y}$ has a nonzero component along h .

More precisely, this exact hyperplane is a maximal totally isotropic subspace for the original Hermitian current form:

$$\mathcal{H}_N = \text{span}_{\mathbb{C}}\{h\} \oplus \ker W, \quad \langle z, Wz' \rangle = 0 \quad (z, z' \in \mathcal{H}_N), \quad \dim_{\mathbb{C}} \mathcal{H}_N = \dim B - 1. \quad (\text{TR12a})$$

The current matrix TR13 has zero h, h entry and is zero on the orthogonal complement of g, h , proving the stated vanishing of the complete restricted form. Its two nonzero eigenvalues have opposite signs, so the induced nondegenerate form on $K^\perp / \ker W$ has signature $(1, 1)$. A totally isotropic subspace of that two-dimensional quotient has dimension at most one: dimension two would be the whole space and would force the nondegenerate form to vanish. Every totally isotropic subspace of the original space therefore has dimension at most $1 + \dim \ker W = \dim B - 1$, attained by $\mathcal{H}_N$. Its quotient $\mathcal{H}_N / \ker W$ is the exactly specified isotropic line generated by the class of h . The operator itself does not vanish on h : TR13 gives $Wh = -i\epsilon\sqrt{\delta}g \neq 0$. Thus the space records vanishing of the restricted current form, while retaining the nonzero action into its complementary direction.

3. Canonical phases in the two current eigenspaces

In $K^\perp$, the original centered Hermitian current is

$$W = i\epsilon(vu^\dagger - uv^\dagger).$$

Its exact matrix in the retained pair g, h is

$$W|_{\text{span}\{g, h\}} = \epsilon \begin{pmatrix} -2\Im r & -i\sqrt{\mathfrak{d}} \\ i\sqrt{\mathfrak{d}} & 0 \end{pmatrix}, \quad Wy_0 = 0. \quad (\text{TR13})$$

Substitute $u = \sqrt{a}g$ and $v = (r/\sqrt{a})g + \sqrt{\mathfrak{d}/a}h$ to prove this directly. Thus

$$\lambda_+ = \epsilon[-\Im r + \sqrt{\mathfrak{d} + (\Im r)^2}] > 0, \quad \lambda_- = \epsilon[-\Im r - \sqrt{\mathfrak{d} + (\Im r)^2}] < 0.$$

Put $\lambda_p = \lambda_+$, $\lambda_n = -\lambda_-$, and

$$A_* = \sqrt{\frac{\lambda_p}{\lambda_p + \lambda_n}}, \quad B_* = \sqrt{\frac{\lambda_n}{\lambda_p + \lambda_n}}, \quad w_+ = -iA_*g + B_*h, \quad w_- = iB_*g + A_*h. \quad (\text{TR14})$$

These are an orthonormal eigenpair for W : the characteristic equation is $\lambda^2 + 2\epsilon(\Im r)\lambda - \epsilon^2\mathfrak{d} = 0$, so substituting TR14 in TR13 verifies both equations. Their phases are fixed by positive real h -components. This definition uses the original real- y monic phases in TR4 and the exact Gram map TR9; it is not an arbitrary identification of two eigenvector phases.

Write

$$z_+ = \langle w_+, \hat{y} \rangle = iA_*c_g + B_*c_h, \quad z_- = \langle w_-, \hat{y} \rangle = -iB_*c_g + A_*c_h, \quad p_\pm = |z_\pm|^2. \quad (\text{TR15})$$

The two exact complex amplitude identities are

$$\boxed{A_*z_+ - B_*z_- = ic_g, \quad B_*z_+ + A_*z_- = c_h.} \quad (\text{TR16})$$

They follow from $A_*^2 + B_*^2 = 1$. In particular TR10 proves

$$|A_*z_+ - B_*z_-| \leq \eta_N.$$

Unlike a difference of squared magnitudes, this inequality controls the relative complex phase. It also remains valid when one or both amplitudes vanish.

Let

$$\delta_N^{\text{tr}} = \frac{|\Im r|}{\sqrt{\mathfrak{d} + (\Im r)^2}} \leq \sqrt{\alpha/\Theta_N}.$$

Since

$$|A_* - B_*| = \frac{|A_*^2 - B_*^2|}{A_* + B_*} = \frac{\delta_N^{\text{tr}}}{A_* + B_*} \leq \delta_N^{\text{tr}}, \quad A_* + B_* \leq \sqrt{2},$$

TR15 gives the finite, unweighted coherent-amplitude estimate

$$\boxed{|z_+ - z_-| \leq \sqrt{2}\eta_N + \delta_N^{\text{tr}}.} \quad (\text{TR17})$$

Here $|c_h| \leq 1$ is enough; no positive lower bound on the active current-plane mass is assumed.

The exact identity

$$p_+ + p_- - 2\Re(z_+\overline{z_-}) = |z_+ - z_-|^2$$

and TR11 yield the completed stronger receiver

$$\boxed{0 \leq p_+ + p_- - 2\Re(z_+\overline{z_-}) \leq (\sqrt{2}\eta_N + \delta_N^{\text{tr}})^2 \leq \exp[-2q\psi(t_N) + O_{h,\omega}(k \log q)].} \quad (\text{TR18})$$

Thus the real part of the sign-sector coherence equals half their combined weight up to a nonnegative error of the doubled exponential order. The phase $\arg(z_+/z_-)$ is not defined at zero amplitude; TR18 is the globally valid statement. It strengthens RC44, which bounded only $|p_+ - p_-|$ at the single exponential order.

For clarity, the preceding weight estimate follows from this stronger one: since $|z_+ + z_-| \leq \sqrt{2(p_+ + p_-)} \leq \sqrt{2}$,

$$|p_+ - p_-| = |\Re((z_+ - z_-)\overline{(z_+ + z_-)})| \leq \sqrt{2}|z_+ - z_-| \leq 2\eta_N + \sqrt{2}\delta_N^{\text{tr}}. \quad (\text{TR19})$$

This derivation preserves all zero directions and does not divide by either sign-sector weight.

4. A quantitatively isolated residual current

The normalized current on the original terminal class is

$$j_N = \frac{\Phi_N(\Lambda x)}{\|\Lambda x\|_{Q_B}^2} = \langle \hat{y}, W \hat{y} \rangle.$$

Using TR13 gives the exact expression and error estimate

$$j_N = 2\epsilon\sqrt{\delta}\Im(\bar{c}_g c_h) - 2\epsilon(\Im r)|c_g|^2,$$

$$\left| j_N - 2\epsilon\sqrt{\delta}\Im(\bar{c}_g c_h) \right| \leq 2\epsilon|r|\eta_N^2 \leq \exp[-2q\psi(t_N) + O_{h,\varpi}(k \log q)]. \quad (\text{TR20})$$

Indeed $\epsilon = e^{q\psi + O(k \log q)}$, $|r| \leq \sqrt{\alpha} \leq e^{-q\psi + O(k \log q)}$, and $\eta_N^2 \leq e^{-2q\psi + O(k \log q)}$. The diagonal trace correction is therefore absolutely exponentially small. The leading term can remain subexponential: multiplication by ϵ reverses the single exponential size of c_g . Neither TR18 nor TR20 evaluates the sign of this retained imaginary product. It is an actual original-metric pairing, with its exact scale and negligible trace term now determined.

All these estimates hold separately at $N = q - 1, q, 2q - 1, 2q$, whose unchanged arguments are $0, 1/q, 1, 1 + 1/q$. No individual sign or four-cutoff signed-current value is obtained by treating the positive upper bounds as signed values.

5. Independent check of the response-amplification dictionary

The source physical convention is $b_n^S = i^n b_n$. Thus, with $F_{13} = \langle b_N^S, x \rangle$, $F_{23} = \langle b_{N+1}^S, x \rangle$, one has

$$F_{13} = i^{-N} \sqrt{E_N} A, \quad F_{23} = i^{-(N+1)} \sqrt{F_N} B, \quad \mathfrak{c}_{N,\lambda} = \frac{\overline{F_{13} F_{23}}}{\omega_N \|x\|^2} = -i\epsilon \frac{\bar{A}B}{\|x\|^2}. \quad (\text{TR21})$$

This verifies the imaginary factor in the real- y coordinate. NCE9-13 prove for this exact terminal eigenclass $\Re \mathfrak{c}_{N,\lambda} = k\delta$ and $|\Im \mathfrak{c}_{N,\lambda}| < k\gamma$. In particular

$$k\delta \leq |\mathfrak{c}_{N,\lambda}| < |\omega|.$$

Both denominators are nonzero. Define the original dimensionless responses

$$\mathfrak{A} = \frac{\|x\|^2}{|A|^2}, \quad \mathfrak{B} = \frac{\|x\|^2}{|B|^2}.$$

The exact phase dictionary TR21 and the eigenclass estimate TR7 prove

$$\mathfrak{A}\mathfrak{B} = \frac{\epsilon^2}{|\mathfrak{c}_{N,\lambda}|^2}, \quad \frac{\epsilon^2}{D^2} \leq \mathfrak{A} \leq \frac{\epsilon^2}{(k\delta)^2}, \quad 1 \leq \mathfrak{B} \leq \frac{D^2}{|\mathfrak{c}_{N,\lambda}|^2} \leq \frac{D^2}{(k\delta)^2}. \quad (\text{TR22})$$

For the upper bound on $\mathfrak{A}$, use $\mathfrak{B} \geq 1$, which is Cauchy-Schwarz with the original unit vector f . For its lower bound use TR7. Then divide the exact product by that lower bound to bound $\mathfrak{B}$. The precise source estimate $\log(u_0/\ell_0) = O_h(k + \log q)$ gives

$$\log \mathfrak{B} = O_h(k + \log q), \quad \log \mathfrak{A} = 2q\psi(t_N) + O_{h,\varpi}(k \log q). \quad (\text{TR23})$$

These independently verify the root's separate leading amplification rates.

If $\beta_K = \|(I - P)f\|^2$, the original response ratios NCE14 are

$$U = \frac{\langle e, (I - P)x \rangle}{A}, \quad V = \frac{\langle f, (I - P)x \rangle}{B}.$$

The physical unit factors in TR21 cancel in each respective ratio. Cauchy-Schwarz on the two actual hidden vectors and TR22 yield

$$|U| \leq \sqrt{\mathfrak{A}\alpha\theta_K} \leq \frac{\epsilon\sqrt{\alpha\theta_K}}{|\mathfrak{c}_{N,\lambda}|}, \quad |V| \leq \sqrt{\mathfrak{B}\beta_K\theta_K} \leq \frac{D\sqrt{\beta_K\theta_K}}{|\mathfrak{c}_{N,\lambda}|}. \quad (\text{TR24})$$

The first displayed upper bound for $\mathfrak{A}$ in TR22 was deliberately weaker than its exact $\epsilon^2/|c|^2$ upper bound; using the latter proves TR24. No individual response or its sign is inferred from these magnitude bounds.

6. Proof provenance and remaining calculation

The exact original action, metric isometry and evolution are proved in CE1–8 of the sealed 026 EVOLUTION_PROOF.md. The terminal conductor coefficient, reverse lift and norm comparison are RC35–42, with the exact period domain at RC37–39. RC25 supplies the complete source rates used in TR11, TR18, TR20 and TR23. These sealed files remain unchanged; this derivation is a new successor.

The earlier underlying proofs are public in the pinned source bank:

- OB1–15: complete terminal observation comparison.
- OC50–74: exact original large-period domain.
- NCE10–21: physical source phases, even spectral measure, strict scalar-current interval and response map.
- FW7: complete original source comparison through degree $2q + 2$.

The independent additions here are the complex hyperplane estimate TR10–12, its exact maximal-isotropic geometry TR12a, the canonical sign-eigenbasis calculation TR13–17, the doubled-rate coherence theorem TR18, and the exponentially small absolute trace correction TR20. The finite proof uses only the stated exact original operators and elementary inner-product algebra; no external theorem has been silently used. The inherited source-polynomial and Gamma identities retain the human citations in the complete CE/OCP proofs: T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt (DLMF Chapter 18); R. A. Askey and R. Roy (DLMF Chapter 5); and the elliptic profile B. C. Carlson (DLMF Chapter 19).

What remains is the signed value of $2\epsilon\sqrt{\mathfrak{d}}\Im(\overline{c_g}c_h)$ on the original terminal class and actual period-dependent projection. The proved complex constraint does not set it to zero. Its sign cannot be selected from the exponentially balanced weights, and no RH conclusion is made.